

Ultra Realistic Voice AI

1. Executive Summary:

Project solves the "Cold Call Grind" by automating the initial outreach and screening phases. Unlike traditional IVR bots, Vox uses Ultra-Realistic Latency- Optimized Voice AI to pitch roles, answer candidate questions, and assess intent in real-time. It is built to sound like a native professional peer, not a machine.

2. Technical Architecture & Latency Goals:

To achieve a human-like flow, the Voice-to-Voice (V2V) pipeline must operate with sub-500ms latency. The integration requires a synchronized dance between VAD, STT, LLM, and TTS.

- A) VAD (Detection) - Neural Voice Activity Detection (WebRTC/Silero) - <50ms
- B) STT(Transcription) - Whisper-Large (Quantized) / Deepgram (Indian English Model) - <150ms
- C) LLM (Reasoning) - Groq Llama-3-70B / Quantized local weights - <100ms
- D) TTS (Synthesis) - ElevenLabs / Play.ht (Streaming API) with Indian Accents - <200ms

3. Localized Conversational Logic:

- A) Native Accent Adaptation: The engine must detect the candidate's regional dialect within the first 5 seconds and adjust the TTS inflection to match (e.g., neutral North Indian vs. South Indian English) to build instant rapport.
- B) Bilingual "Hinglish" Support: The LLM must be prompt-engineered to handle seamless code-switching between Hindi and English, ensuring the conversation feels natural to the Bharat workforce.

4. Acceptance Criteria (AC):

- A) Intent Scoring: Post-call, the AI must generate a 1-10 "Interest Score" based on tone and answer quality.
- B) Dynamic FAQ: The AI must be able to answer at least 3 non-scripted questions about the company (e.g., office location, benefits) before returning to the screening script.
- C) Closed-Loop Bridge: Automated trigger of a WhatsApp follow-up containing the interview calendar link immediately upon hanging up.

CRITICALITY:

The #1 "Bot-Giveaway" is over-talking. The team MUST build a sophisticated interruption-handler. If the candidate speaks while the AI is talking, the AI must truncate its audio stream within 100ms and shift back to "Listening" mode immediately.